ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE D · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Practical work is not marked strictly on syntax. Give marks for correct logic, correct pins and connections, sensible comments and neat work. Accept other pin choices and layouts that work.

QUESTION 1

1.1 LED anode to a GPIO pin ✓✓ through the 220 Ω resistor ✓✓; LED cathode to GND ✓; button between a GPIO pin ✓ and GND ✓; GND rail to a Raspberry Pi GND pin ✓; LED polarity correct ✓; neat, no short circuits ✓ (10)

1.2 Start ✓; set up LED and button pins, best time = large value ✓; repeat 3 rounds ✓; LED off and random wait of 2–5 s ✓✓; decision: button pressed during the wait? → "Too early!" ✓; LED on and start time recorded ✓; wait for the button ✓; reaction time = end − start, displayed ✓; decision: faster than best? → best = time ✓; after 3 rounds display the best time and stop ✓ (11)

1.3 Circuit ✓ and flowchart ✓ saved and printed / presented (2)

[23]

QUESTION 2

2.1 File created and named ✓✓ (2)

2.2 (1) PUD_UP ✓ (2) OUT ✓ (3) False ✓ (4) LOW ✓ (5) armed ✓ (6) a short debounce delay, e.g. 0.5 ✓ (7) HIGH ✓ (8) HIGH ✓ (9) LOW ✓ (10) KeyboardInterrupt ✓ (11) cleanup ✓ (11)

2.3 Useful comments, about 1 mark each ✓✓✓✓✓✓✓✓✓✓, e.g. libraries; pin numbers; BCM numbering; door switch input with pull-up; ARM button input; outputs; armed starts as False; button toggles armed; debounce delay; door open while armed switches on the LED and buzzer; otherwise outputs off; Ctrl + C handling; cleanup. (10)

2.4 Saved ✓ and printed / presented ✓ (2)

[25]

QUESTION 3

3.1 LDR module VCC to 3,3 V ✓, GND to GND ✓, DO to a GPIO pin ✓; each white LED with its own 220 Ω resistor to a GPIO pin (or both on one pin through their own resistors) ✓✓; green LED with 220 Ω to a GPIO pin ✓; all cathodes to GND ✓; LED polarities correct ✓; common GND ✓; neat, labelled ✓ (10)

3.2 File created and named ✓✓ (2)

3.3 Libraries imported and pins set up ✓✓; green LED on to show the system is working ✓; reads DO in a loop ✓; dark (HIGH) switches the street lights on ✓✓; light switches them off ✓; the time they switched off is recorded ✓ and they stay off for at least 10 s ✓; try / finally with cleanup ✓, e.g. (10)

```
import RPi.GPIO as GPIO
import time

SENSOR = 4                    # LDR module DO, HIGH = dark
LIGHTS = [17, 27]             # two white LEDs (street lights)
POWER = 22                    # green LED: system working

GPIO.setmode(GPIO.BCM)
GPIO.setup(SENSOR, GPIO.IN)
GPIO.setup(LIGHTS + [POWER], GPIO.OUT)
GPIO.output(POWER, GPIO.HIGH)

lights_on = False
off_time = 0
try:
    while True:
        dark = GPIO.input(SENSOR) == GPIO.HIGH
        if dark and not lights_on and time.time() - off_time >= 10:
            GPIO.output(LIGHTS, GPIO.HIGH)
            lights_on = True
        elif not dark and lights_on:
            GPIO.output(LIGHTS, GPIO.LOW)
            lights_on = False
            off_time = time.time()   # stay off for at least 10 s
        time.sleep(0.5)
except KeyboardInterrupt:
    print("Street light stopped")
finally:
    GPIO.cleanup()
```

3.4 Name and description ✓✓; basic operation – power on, what the green LED means, lights switch on in the dark and off in daylight, 10-second delay ✓✓✓; troubleshooting – at least two problems with solutions (e.g. lights never switch on: check the sensor wiring or turn its sensitivity screw; green LED off: check the power supply) ✓✓✓; at least two sensible FAQs with answers ✓✓ (10)

[32]

TOTAL: 80

END OF MARKING GUIDELINE